

Actual Active Support on an Empty Eligible Family: A Vacuity Firewall for the H1 Carrier Root

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July 2026

Abstract

The H1 occurrence architecture requires an actual active-support certificate independently of a source-backed local occurrence-edge family. TPC-175 and TPC-176 leave no eligible production carrier in the frozen TPC-133–172 source scope. We audit literal physical coefficients only on that eligible domain. No carrier is tested, so the counts of nonzero, zero, and missing coefficients are all zero.

The resulting universal support statement is vacuous and cannot supply an existential witness. We formalize this as a firewall: the empty-domain audit passes, while `H1.actual_active_support_certificate` remains NOT-TESTABLE. The independent H9 literal-weight registry is neither imported nor closed and retains decay axis NONE. This is an L1 interface obstruction, not physical arithmetic progress or program-positive L2.

1 Two independent prerequisites

TPC-166 splits the former occurrence crosswalk monolith into three independent roots [1]. Write

$$\begin{aligned} E &= \text{a source-backed production local edge family,} \\ A &= \text{actual active support of its physical carriers,} \\ M &= \text{a canonical or minimal physical representation.} \end{aligned} \tag{1}$$

Neither $E \Rightarrow A$ nor $A \Rightarrow E$ is available. TPC-177 studies A without collapsing this independence.

For a candidate carrier c , an active-support record must contain at least a literal target-expression locator, the literal physical coefficient $w(c)$, its target normalization, and a proof that

$$w(c) \neq 0. \tag{2}$$

Formal appearance in a schema, archive, completion fiber, or synthetic fixture does not satisfy (2).

2 The eligible domain

TPC-175 certifies no qualifying production local edge in the frozen source inventory, and TPC-176 therefore exposes no eligible carrier:

$$\mathcal{C}_{\text{elig}} = \emptyset. \tag{3}$$

The support audit is deliberately restricted to (3). Adding a formal archived cut or a synthetic carrier would fabricate the missing source-backed edge.

Define

$$\begin{aligned} \mathcal{C}_+ &= \{c \in \mathcal{C}_{\text{elig}} : w(c) \neq 0 \text{ is source-proved}\}, \\ \mathcal{C}_0 &= \{c \in \mathcal{C}_{\text{elig}} : w(c) = 0 \text{ is source-proved}\}, \\ \mathcal{C}_? &= \{c \in \mathcal{C}_{\text{elig}} : \text{no complete coefficient record exists}\}. \end{aligned} \tag{4}$$

These three classes partition the eligible domain when every carrier is assigned exactly one audit outcome.

Proposition 2.1 (Exact coefficient ledger). *For the TPC-176 domain,*

$$(|\mathcal{C}_{\text{elig}}|, |\mathcal{C}_+|, |\mathcal{C}_0|, |\mathcal{C}_?|) = (0, 0, 0, 0). \quad (5)$$

Proof. Immediate from (3) and the definitions in (4). $\square$

3 The vacuity firewall

On the empty domain, the sentence

$$\forall c \in \mathcal{C}_{\text{elig}}, \quad w(c) \neq 0 \quad (6)$$

is true in classical logic. But the H1 root requires production content:

$$\exists c \in \mathcal{C}_{\text{elig}} \quad \text{with a source-backed local edge and } w(c) \neq 0. \quad (7)$$

Theorem 3.1 (TPC-177 vacuity firewall). *Equation (6) does not prove (7). The executable audit status is `VACUOUS_EMPTY_ELIGIBLE_DOMAIN`, and the H1 actual-active-support root remains `NOT-TESTABLE`.*

Proof. An existential witness for (7) would be an element of the empty set in (3), which is impossible within the declared audit instance. This does not prove that an actual active carrier is mathematically impossible; it proves that the current source-locked input supplies none. Hence neither `PROVED` nor a global `STOP_ROUTE` classification is valid. $\square$

The distinction may be summarized as follows.

Statement	Value	Evidential consequence
empty-domain audit internally consistent	true	the ledger and partition checks pass
all eligible carriers active	vacuously true	no existential content
an active production carrier exists	unproved	H1 support root remains <code>NOT-TESTABLE</code>
active carriers do not exist mathematically	unproved	frozen-corpus scope cannot imply nonexistence

4 Separation from the H9 literal-weight registry

The H1 support root and H9.literal_weight_registry have different jobs. H1 asks whether a source-backed occurrence carrier is present with a nonzero physical coefficient. H9 supplies literal weight data to a later arithmetic synthesis. A registry has data semantics and decay axis `NONE`; it does not manufacture a cancellation exponent.

TPC-177 neither scans nor imports the H9 registry. Thus it records

$$\begin{aligned} \text{H1 active-support status} &= \text{NOT_TESTABLE}, \\ \text{H9 literal-weight status} &= \text{NOT_IMPORTED}, \\ \text{decay created by either registry} &= 0. \end{aligned} \quad (8)$$

Even if a future H9 registry names a literal coefficient, an independent source-backed H1 carrier would still be required.

5 Reproducibility and claim boundary

The standard-library audit source-locks the TPC-175 maximal family and TPC-176 coverage ledger, recomputes (3)–(5), and checks the existential firewall. Mutations that fabricate an eligible carrier, promote vacuous truth to an H1 certificate, import H9 implicitly, assign H9 a decay exponent, or promote the result to L2 are rejected. Hashes have integrity semantics only.

No fixed physical $h_0 = 2$, named phase, deterministic endpoint, target normalization, or endpoint slack is consumed. Accordingly there is no strict 1/400 gain, prime-pair lower bound, or twin-prime conclusion.

6 Conclusion

An empty eligible domain makes an audit easy but a theorem impossible: the universal support sentence is vacuous, and the required existential production witness is absent. TPC-177 keeps the H1 support root open and keeps it disjoint from the H9 literal-weight registry. TPC-178 applies the same eligibility discipline to canonical and minimal physical representations.

References

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